(558250-1) Data Analysis and Visualization

Genetic Algorithm

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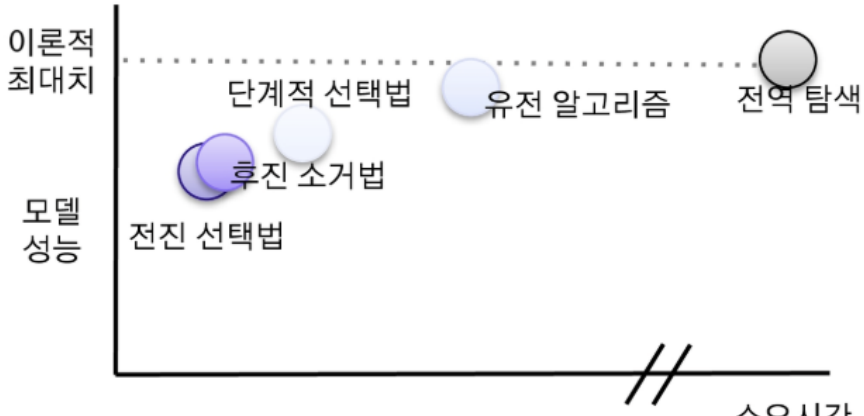
1 Background

Feature Selection Methods

- **Limitations of the previous variable selection methods:**
  - ✓ Exhaustive search: guarantee the optimal subset, but takes too long time (practically impossible for many tasks)
  - ✓ Local search (forward/backward/stepwise): efficient search but the search space is very limited, which leads to a low probability of finding the optimal solution

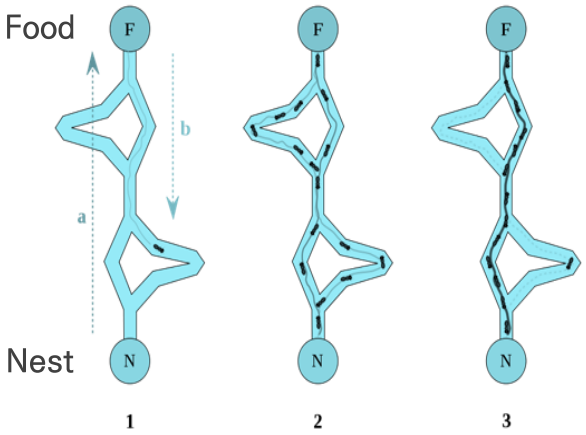
- **Idea**
  - ✓ Improve the performance of local searches with a little additional computational time!



[모델 성능 및 소요 시간]

Meta-Heuristic Approach

- Solve a complex problem by doing trials and errors efficiently
- Among the optimization algorithms, many of them mimic the way of a natural system works



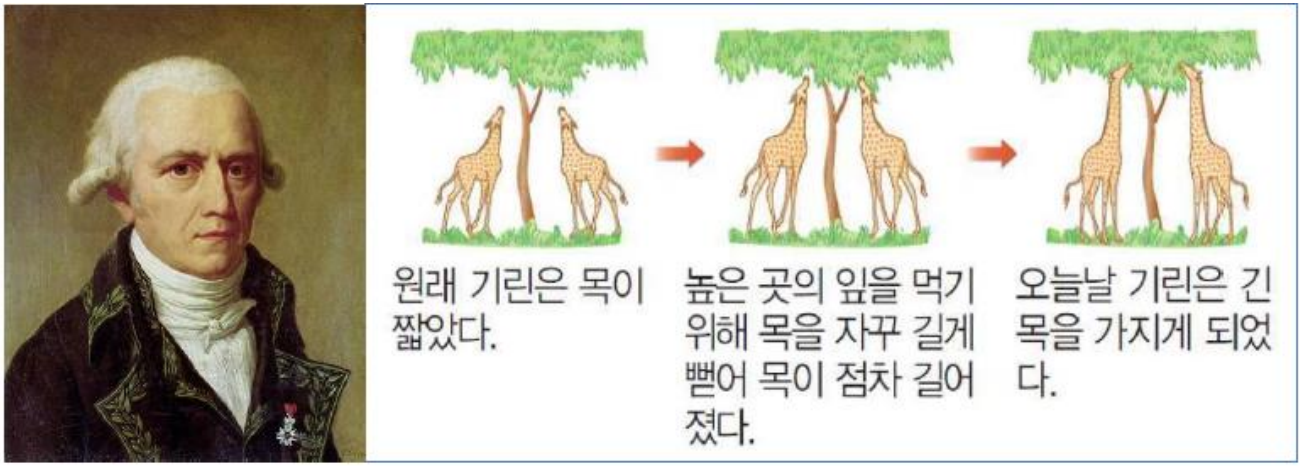
[Ant Colony Algorithm]

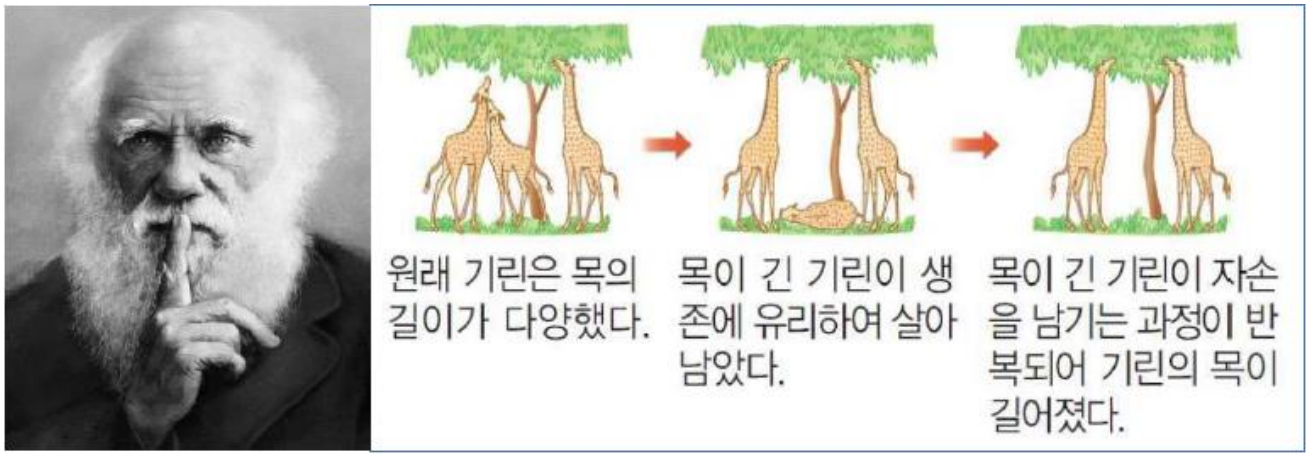
[Particle Swarm Optimization]

[Simulated Annealing]

Meta-Heuristic Approach

- 라마르크 용불용설 vs. 다윈 자연선택설

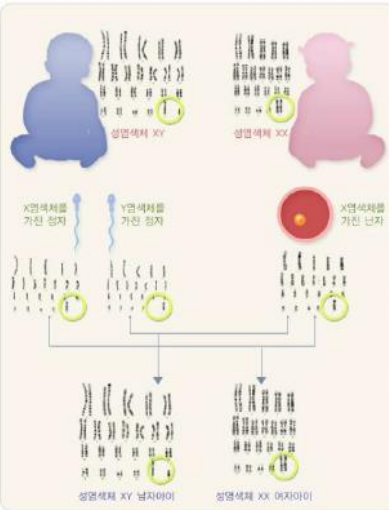


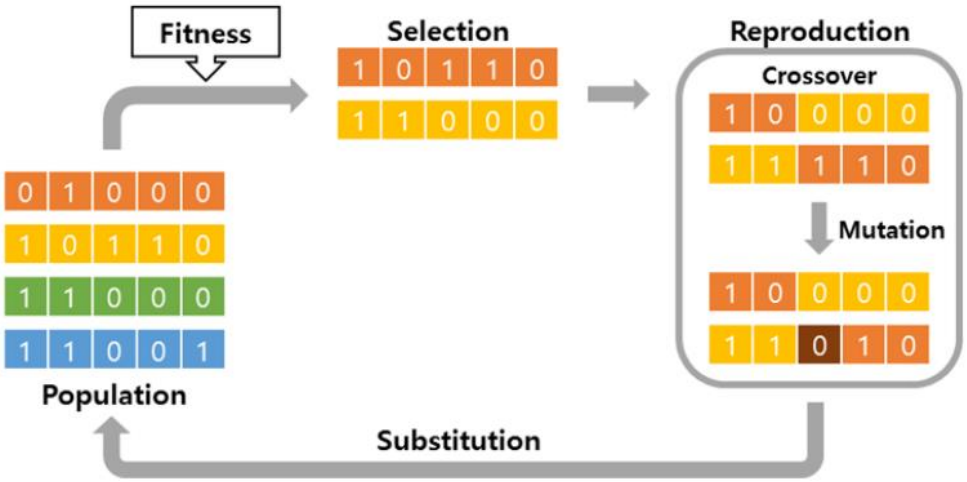


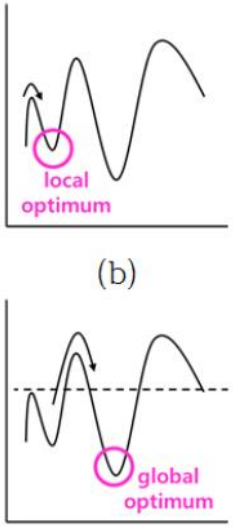
2 Genetic Algorithm

Genetic Algorithm

- An evolutionary algorithm that mimics the reproduction of creatures
- Find a superior solutions and preserve by repeating the reproduction process
  - ✓ Selection: Select a superior solution to improve the quality
  - ✓ Crossover: Search various alternatives based on the current solutions
  - ✓ Mutation: Give a chance to escape the local optimum

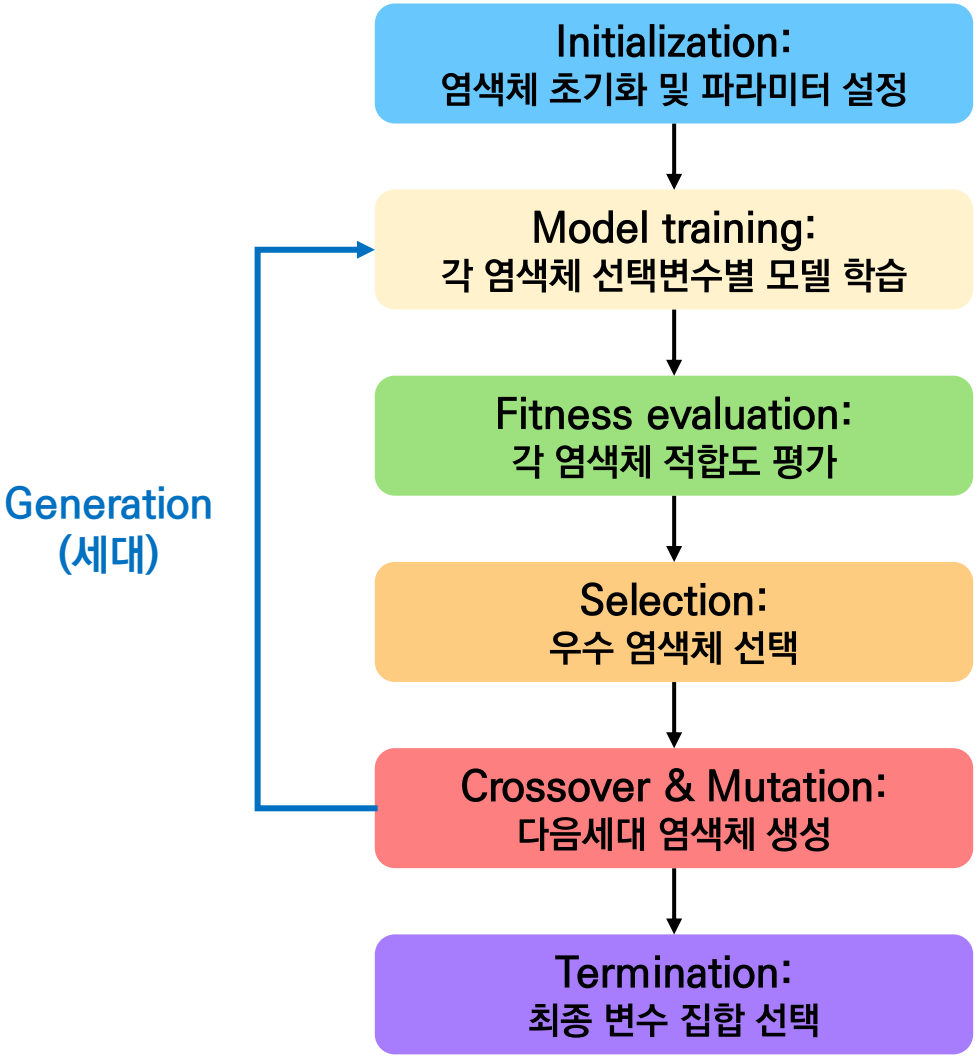






Genetic Algorithm for Feature Selection

- How genetic algorithm works:



GA Step 1: Initialization

- **Encoding chromosomes**
  - ✓ Genetic algorithm can be used not only for variable selection, but for a wide range of optimization problems
  - ✓ Encoding scheme can be different for different tasks
  - ✓ Binary encoding is commonly used for variable selection

Chromosome (염색체)

Gene (유전자)

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	…	x_d
1	0	0	1	0	1	1	0	…	1

1: Use the corresponding variable in the modeling

0: Do not use the variable

GA Step 1: Initialization

- **Parameter Initialization**
  - ✓ The number of chromosome (population)
  - ✓ Fitness function
  - ✓ Crossover mechanism
  - ✓ The rate of mutation
  - ✓ Stopping criteria: minimum fitness improvement, maximum iterations, etc.

GA Step 1: Initialization

- **Example: Population Initialization**
  - ✓ Random number generation for each gene
  - ✓ Convert the random numbers to binary values (cut-off = 0.5 in this example)

	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Chromosome 1	0.16	0.74	0.90	0.96	0.19	0.70	0.31	0.23	0.83	0.62
Chromosome 2	0.71	0.75	0.82	0.83	0.41	0.97	0.14	0.40	0.89	0.66
Chromosome 3	0.36	0.56	0.40	0.08	0.80	0.32	0.44	0.27	0.34	0.31
Chromosome 4	0.30	0.26	0.83	0.71	0.70	0.74	0.11	0.57	0.81	0.01
Chromosome 5	0.27	0.29	0.67	0.54	0.38	0.09	0.07	0.90	0.00	0.53
Chromosome 6	0.70	0.59	0.62	0.18	0.36	0.02	0.45	0.24	0.40	0.06
Chromosome 7	0.43	0.49	0.34	0.36	0.26	0.67	0.09	0.68	0.29	0.44
Chromosome 8	0.10	0.96	0.20	0.16	0.65	0.07	0.73	0.14	0.91	0.77

	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Chromosome 1	0	1	1	1	0	1	0	0	1	1
Chromosome 2	1	1	1	1	0	1	0	0	1	1
Chromosome 3	0	1	0	0	1	0	0	0	0	0
Chromosome 4	0	0	1	1	1	1	0	1	1	0
Chromosome 5	0	0	1	1	0	0	0	1	0	1
Chromosome 6	1	1	1	0	0	0	0	0	0	0
Chromosome 7	0	0	0	0	0	1	0	1	0	0
Chromosome 8	0	1	0	0	1	0	1	0	1	1

GA Step 2: Training

- **Example: Train the model**
  - ✓ Assume that the model is a multiple linear regression (MLR)

	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Chromosome 1	0	1	1	1	0	1	0	0	1	1
Chromosome 2	1	1	1	1	0	1	0	0	1	1
Chromosome 3	0	1	0	0	1	0	0	0	0	0
Chromosome 4	0	0	1	1	1	1	0	1	1	0
Chromosome 5	0	0	1	1	0	0	0	1	0	1
Chromosome 6	1	1	1	0	0	0	0	0	0	0
Chromosome 7	0	0	0	0	0	1	0	1	0	0
Chromosome 8	0	1	0	0	1	0	1	0	1	1

For chromosome 1, fit the MLR model

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3 + \hat{\beta}_4 x_4 + \hat{\beta}_6 x_6 + \hat{\beta}_9 x_9 + \hat{\beta}_{10} x_{10}$$

For chromosome 2, fit the MLR model

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3 + \hat{\beta}_4 x_4 + \hat{\beta}_6 x_6 + \hat{\beta}_9 x_9 + \hat{\beta}_{10} x_{10}$$

And so on…

GA Step 3: Fitness Evaluation

- Fitness function is a criterion that determines which chromosomes are better than others.
- In general, the higher the fitness value, the better the chromosomes.
- Common criteria that are embedded in the fitness function.
  - ✓ If two chromosomes have the same fitness value, the one with fewer variables is preferred
  - ✓ If two chromosomes use the same number of variables, the one with higher predictive performance is preferred
- In case of multiple linear regression:
  - ✓ R^2 (coefficient of determination)
  - ✓ Adjusted R^2
  - ✓ …

GA Step 3: Fitness Evaluation

- Example: Fitness function

	Adj R^2	Rank	Weight
Chromosome 1	0.75	2	0.177
Chromosome 2	0.78	1	0.184
Chromosome 3	0.50	5	0.118
Chromosome 4	0.65	3	0.154
Chromosome 5	0.40	6	0.095
Chromosome 6	0.35	7	0.083
Chromosome 7	0.25	8	0.059
Chromosome 8	0.55	4	0.130

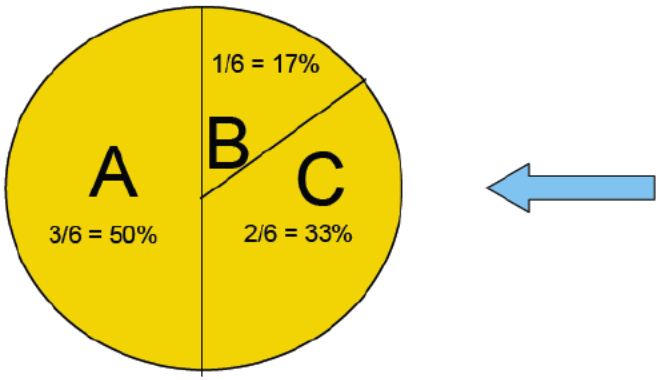
✓ Adjusted R^2 is used for the fitness function

✓ Each chromosome's weight is its Adj. R^2 divided by the sum of Adj. R^2 s of all chromosomes

⇒ Used for probabilistic chromosome selection

GA Step 4: Selection

- Select superior chromosomes in the current population to reproduce the population of the next generation.
- Deterministic selection
  - ✓ Select only top N% of chromosomes
  - ✓ Bottom (100−N)% chromosomes are never selected
- Probabilistic selection
  - ✓ Use the fitness value of each chromosome as the selection weight
  - ✓ All chromosomes can be selected with different probabilities



fitness(A) = 3

fitness(B) = 1

fitness(C) = 2

GA Step 4: Selection

- **Example: Selection**

✓ Case 1: Deterministic selection

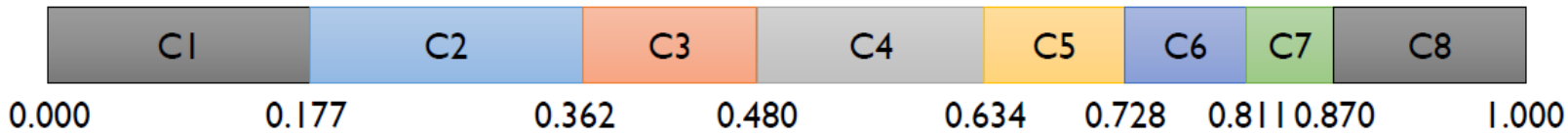
	Adj R^2	Rank	Weight
Chromosome 1	0.75	2	0.177
Chromosome 2	0.78	1	0.184
Chromosome 3	0.50	5	0.118
Chromosome 4	0.65	3	0.154
Chromosome 5	0.40	6	0.095
Chromosome 6	0.35	7	0.083
Chromosome 7	0.25	8	0.059
Chromosome 8	0.55	4	0.130

✓ Only the chromosomes 1, 2, 4, and 8 can be a parent for the next generation

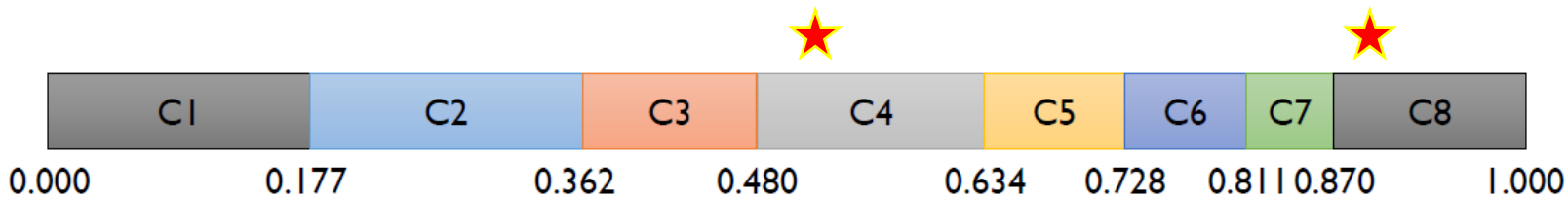
	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Chromosome 1	0	1	1	1	0	1	0	0	1	1
Chromosome 2	1	1	1	1	0	1	0	0	1	1
Chromosome 4	0	0	1	1	1	1	0	1	1	0
Chromosome 8	0	1	0	0	1	0	1	0	1	1

GA Step 4: Selection

- **Example: Selection**
  - ✓ Case 1: Probabilistic selection
    - Cumulative weight



- For the first set of parents, generate random numbers: (0.881, 0.499)

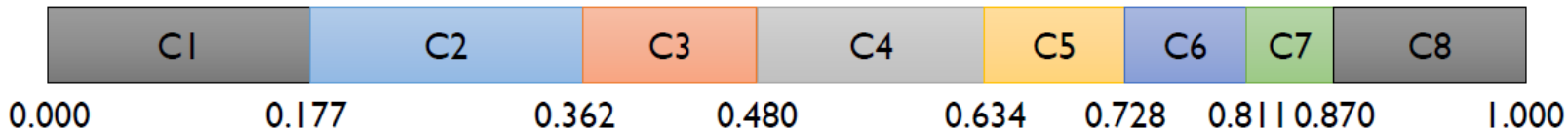


- Choose the chromosome 4 and 8

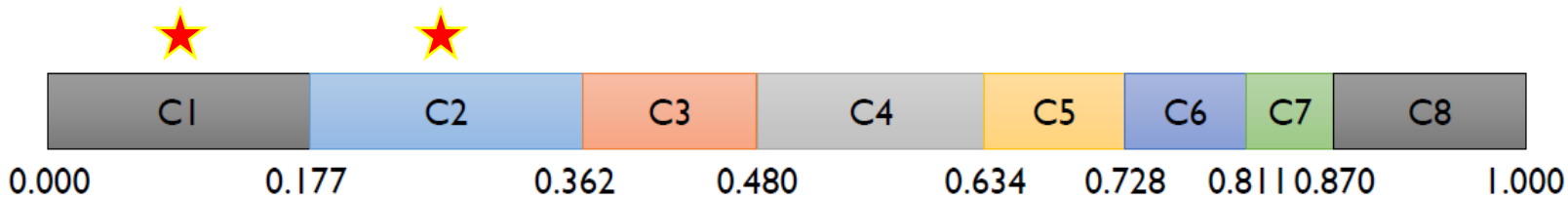
	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Chromosome 4	0	0	1	1	1	1	0	1	1	0
Chromosome 8	0	1	0	0	1	0	1	0	1	1

GA Step 4: Selection

- **Example: Selection**
  - ✓ Case 1: Probabilistic selection
    - Cumulative weight



- For the second set of parents, generate random numbers: (0.098, 0.252)



- Choose the chromosome 1 and 2

	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Chromosome 1	0	1	1	1	0	1	0	0	1	1
Chromosome 2	1	1	1	1	0	1	0	0	1	1

GA Step 5: Crossover & Mutation

- **Crossover (Reproduction)**
  - ✓ Two child chromosomes are produced from two parent chromosomes
  - ✓ The number of crossover points can vary from 1 to n (total number of genes)
  - ✓ Example with 1 crossover point

	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Chromosome 4	0	0	1	1	1	1	0	1	1	0
Chromosome 8	0	1	0	0	1	0	1	0	1	1

	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Child 1	0	1	0	0	1	1	0	1	1	0
Child 2	0	0	1	1	1	0	1	0	1	1

GA Step 5: Crossover & Mutation

- **Crossover (Reproduction)**
  - ✓ Two child chromosomes are produced from two parent chromosomes
  - ✓ The number of crossover points can vary from 1 to n (total number of genes)
  - ✓ Example with 2 crossover points

	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Chromosome 4	0	0	1	1	1	1	0	1	1	0
Chromosome 8	0	1	0	0	1	0	1	0	1	1

	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Child 1	0	1	0	1	1	1	0	0	1	1
Child 2	0	0	1	0	1	0	1	1	1	0

GA Step 5: Crossover & Mutation

- **Crossover (Reproduction)**
  - ✓ Two child chromosomes are produced from two parent chromosomes
  - ✓ The number of crossover points can vary from 1 to n (total number of genes)
  - ✓ Example with 10 crossover points

	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Chromosome 4	0	0	1	1	1	1	0	1	1	0
Chromosome 8	0	1	0	0	1	0	1	0	1	1
Random number	0.94	0.89	0.27	0.76	0.54	0.5	0.56	0.08	0	0.46

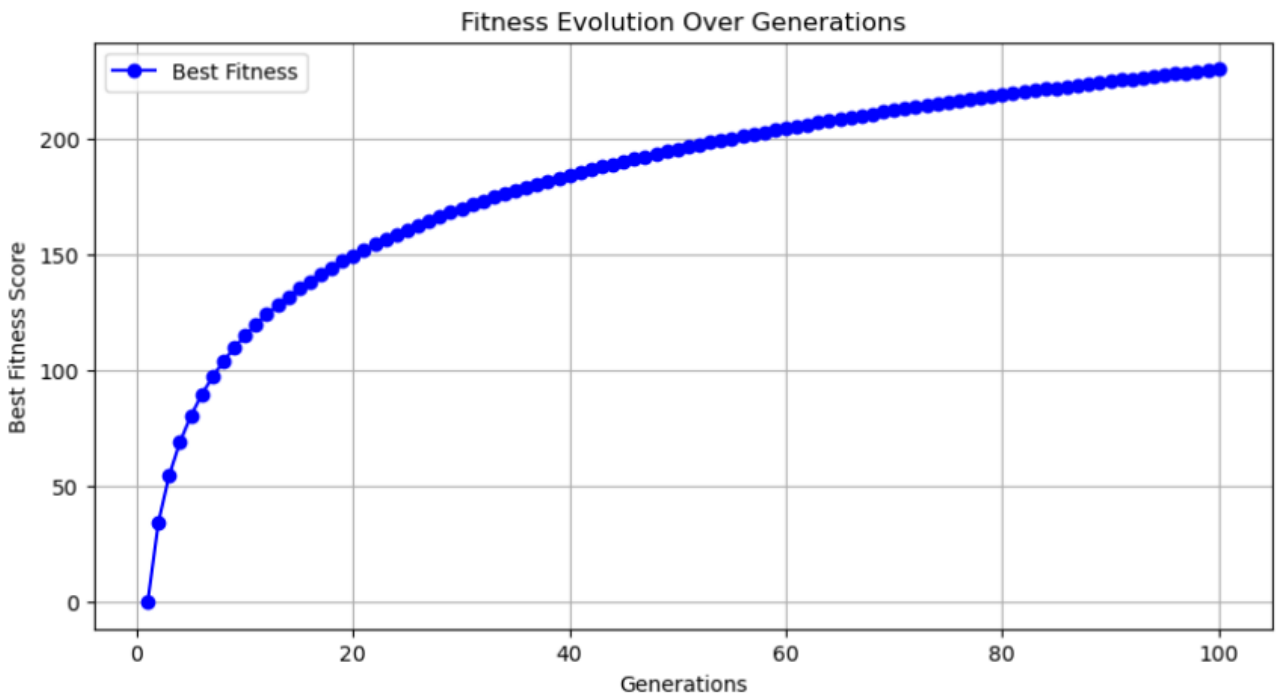
	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Child 1	0	1	1	0	1	0	1	1	1	0
Child 2	0	0	0	1	1	1	0	0	1	1

GA Step 5: Crossover & Mutation

- **Mutation**
  - ✓ Maintain diversity from one generation of a population of chromosomes to the next
  - ✓ Alter one or more gene values in a chromosome from its initial state, which result in entirely new gene values being added to the gene pool
  - ✓ By mutation, the current solution can have a chance to escape from the local optima
  - ✓ A too large mutation rate can increase the time to converge (0.01 can be a good choice)

	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Child 1	0	1	1	0	1	0	1	1	1	0
Random number	0.43	0.35	0.71	0.54	0.62	0.73	0.71	0.92	0.95	0.91
Child 2	0	0	0	1	1	1	0	0	1	1
Random number	0.91	0.03	0.22	0.96	0.32	0.73	0.43	0.32	0.01	0.04

	x1	x2	x3	x4	x5	x6	x7	x8	x9	x10
Child 1	0	1	1	0	1	0	1	1	1	0
Child 2	0	0	0	1	1	1	0	0	0	1

GA Step 6: Termination

- **Find the best solution**
  - ✓ Select the chromosome with the highest fitness value after the stopping criteria are satisfied
  - ✓ Generally, significant fitness improvement occurs in the early stages, which becomes marginal after some generations